THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** A is a better conductor	Faster fall means faster conduction.
2	**(a)** reversed	Land breeze.
3	**(d)** sea breeze	Fluid motion.
4	**(d)** it absorbs more radiation	Dark absorbs.
5	**(d)** nearly stops	Temperatures equalise.
6	**(b)** heating iron rod	Solid transfer.
7	**(a)** hot air rising pushes it	Convection currents.
8	**(c)** not form	No temperature difference.
9	**(d)** solar cooker	Absorbs sunlight.
10	**(c)** radiation	Direct radiation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heating from the top gives little convection (1); From the bottom, hot liquid rises and cooler sinks (1).	1+1
12	Evaporation takes heat (1); Keeps water cool (1).	1+1
13	Blanket is an insulator (1); Slows heat entering (1).	1+1
14	Metal conducts heat to the fins (1); Black radiates heat well (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Thermocol best insulator (1); Ceramic next (1); Metal loses most heat (1).	1+1+1
16	Steel conducts (1); Plastic does not (1); Steel handle becomes hot (1).	1+1+1
17	Trap air (1); Insulation (1); Body heat is retained (1).	1+1+1
	OR: Hot air rises above (1); Small windows reduce radiation (1); Thick walls slow conduction (1).	1+1+1
18	Coolant absorbs heat inside (1); Releases it outside (1); Back coils warm (1).	1+1+1
	OR: Metal handle warms quickly (1); Wooden handle remains cooler (1); Conductivity differs (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Same volume and initial temperature (1); Wrap each beaker (1); Measure every 5 min (1); Same room conditions (1); Wool keeps heat longest (1).	1+1+1+1+1
	OR: Black in sun highest (1); White in sun lower (1); Both in shade low (1); Differences show absorption (1); Conclude (1).	1+1+1+1+1
20	Reflector concentrates light (1); Black inner surface absorbs (1); Glass lid traps heat (1); Insulated box reduces loss (1); Food cooks slowly (1).	1+1+1+1+1
	OR: Wool traps air (1); Cotton lets air pass (1); Insulated lunch boxes (1); Fridges (1); Reflective foil (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** heats and cools quickly	Low heat capacity.
2	**(a)** hot gases rise and draw fresh air	Draught by convection.
3	**(a)** white reflects most radiation	Less absorption.
4	**(b)** water cools slowly	Slow cooling.
5	**(b)** hot air can escape	Hot air rises.
6	**(c)** metal conducts heat away from the body faster	Rate of heat loss.
7	**(d)** they reduce loss of radiation	Trap heat.
8	**(c)** higher than near the floor	Hot air rises.
9	**(d)** the same temperature	Same temperature.
10	**(a)** heat radiates out unhindered	Radiation escapes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Black surface absorbs heat from flame (1); Improves heating (1).	1+1
12	Glass conducts heat better (1); Takes heat from the hand faster (1).	1+1
13	Sand heats and cools quickly (1); Clear sky lets heat radiate out (1).	1+1
14	Reflect radiant heat (1); Protect from burns (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Use a lamp inside a bell jar (1); Sense warmth outside (1); Heat crosses by radiation (1).	1+1+1
16	Stops spread of fire (1); Reduces fuel (1); Controls radiation and convection (1).	1+1+1
17	Hot water rises (1); Convection currents heat all (1); Faster heating (1).	1+1+1
	OR: Day sea breeze cools (1); Night land breeze (1); Moderate temperature (1).	1+1+1
18	Radiation heats (1); Evaporation increases (1); Wind helps (1).	1+1+1
	OR: Radiation passes through glass (1); Seats absorb and heat (1); Hot air trapped (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Flame to base: radiation and convection (1); Base conducts (1); Water: convection (1); Surface loses heat (1); Steam leaves (1).	1+1+1+1+1
	OR: Metal: fast (1); Ceramic: slow, even (1); Retains heat (1); Metal needs lid (1); Choice by recipe (1).	1+1+1+1+1
20	Radiation reaches the Earth (1); Land heats faster (1); Sea slower (1); Pressure differences (1); Winds blow (1).	1+1+1+1+1
	OR: Air gap between panes (1); Poor conductor (1); Less heat loss (1); Reduces noise (1); Saves energy (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** cork and plastic	Both insulators.
2	**(a)** conduction	Contact.
3	**(a)** feeling warmth from a lit bulb in a closed glass jar	Through vacuum or glass.
4	**(d)** heat it from below	Convection.
5	**(d)** convection currents	Hot rises, cold sinks.
6	**(b)** convection and radiation	Air movement and radiation.
7	**(c)** convection needs a fluid	True.
8	**(b)** radiation	Vacuum cannot stop radiation.
9	**(a)** iron conducts heat to his hand	Poor insulator.
10	**(c)** warm air rises and spreads	Convection.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Warm air carries it up (1); Convection (1).	1+1
12	Metal absorbs radiation (1); Conducts heat inside (1).	1+1
13	Reflect sunlight (1); Reduce indoor temperature (1).	1+1
14	Metal conducts heat (1); Wood insulates the hand (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Snow traps air (1); Small entrance retains heat (1); Body heat warms inside (1).	1+1+1
16	No direct radiation (1); Shade reduces heating (1); Less conduction (1).	1+1+1
17	Convection currents (1); Air moves (1); Warm air rises (1).	1+1+1
	OR: Steam raises pressure (1); Higher boiling point (1); Faster cooking (1).	1+1+1
18	Warm air goes out at the top (1); Cold air enters at the bottom (1); Convection currents (1).	1+1+1
	OR: Increase heat loss (1); Moving air cools (1); Blood vessels near the surface (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Kettle body conducts (1); Handle insulates (1); Cooler box needs insulation (1); Foam traps air (1); Purposes differ (1).	1+1+1+1+1
	OR: Rod: conduction (1); Water: convection (1); Sun-Earth: radiation (1); Particles vibrate in rod (1); Fluid moves in water (1).	1+1+1+1+1
20	Place a lamp behind glass (1); Feel warmth through glass (1); Glass does not conduct quickly (1); Touching the glass later (1); Different modes (1).	1+1+1+1+1
	OR: Insulate roof (1); Use curtains (1); Close doors (1); Use lids while cooking (1); Solar heaters (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.